



RIO Source Edition Protocol For Third Party Integrators

Revision 1.16.04

Document History

Revision	Date	Author	Version	Description
1.00.00	9/9/2012	be	v1.0	Initial Release
1.07.00	1/10/13	Pfcass	V1.7	Add Keycode table Add all v1.7 features: - Banks and Presets - MMRate - MMContextMenu - MM Button - 'X' hidden attribute - MMUseForms (and associated notifications) - MMFormEntry
1.07.01	04/03/13	Pfcass	V1.7	Copy section in MM Session Initialization from main manual Remove useless table of contents. It apparently did not convert properly from the original .pages document and will require reformatting of this document to work properly.
1.07.03	7/1/13	Bill R	V1.7	Changed references to MMNextItems and MMPrevItems to KeyRelease PageUp and KeyRelease PageDown for MM Info Screen Removed reference to MMStartChar. This will be available in the RIO 1.9 specification. Removed unsupported feature.
1.08.00	10/20/14	Bill R	V1.8	Corrected repeatMode responses. - Original text indicated valid responses were OFF, SINGLE, ON. - Corrected responses to read OFF, SINGLE, ALL. Corrected notation for shuffleMode. - Notation now excludes iBridge.
1.08.01	12/26/14	Bill R	V1.8	Corrected descriptions for Context Menu and Likes/Dislikes. Presets: - Moved Managing Presets after events section. - Preset events to events table.

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1.10.01	2/13/17	Ed H	<i>V1.10</i>	Media Service Modes and Key Support table omitted artistName key for TuneIn and vTuner. Table was corrected to reflect support of that key for those services.
1.11.00	7/10/17	Ed H	<i>V1.11</i>	Added sections for MMSelectOption and MMKeepAlive Deprecated MMStartChar – use MMStartItem
1.12.00	8/14/17	Ed H	<i>V1.12</i>	Added System.Support to determine capability for managing device grouping and system alarm.
1.14.00	4/17/19	Bill R Ed H	<i>V1.14</i>	Added new display item attributes for MBX series products. Added playStatus GET and notification descriptions Added availableControls GET and notification descriptions Added sampleRate GET and notification descriptions Added bitRate GET and notification descriptions Added bitDepth GET and notification descriptions Added playTime GET and notification descriptions Added trackTime GET and notification descriptions Added SetSeekTime EVENT description Added menu item context menu via MMSelectItem Updated JSON menu items to include value, imgUrl, uri, and attributes fields. Included new example of JSON menu list with additional fields.
1.15.00	10/3/19	Ed H	<i>V1.15</i>	Corrected the list of supported devices for GET S[s].type RIO-SE 1.15.00 does not contain any additions to the API. Version was updated to match RIO.

1.15.01	10/21/19	Bill R Ed H	<i>V1.15</i>	Added GET S[s].Support.MM.restoreOption Added N S[s].Support.restoreOption Added GET S[s].Support.availableControls Added N S[s].Support.availableControls Added GET S[s].Support.trackTime Added N S[s].Support.trackTime Added GET S[s].Support.playTime Added N S[s].Support.playTime Added GET S[s].Support.playerData Added N S[s].Support.playerData Added GET S[s].Support.favoritesV2 Added N S[s].Support.favoritesV2
1.15.02	12/23/19	Bill R Ed H	<i>V1.15</i>	Added user login attribute. See Menu Item Attributes .
1.15.03	2/19/20	Bill R Ed H	<i>V1.15</i>	Added Media Streamer Mode Metadata Support Table
1.15.04	2/19/20	Bill R Ed H	<i>V1.15</i>	Added recommended metadata field usage to Media Streamer Mode Metadata Support Table
1.16.01 1.16.02 1.16.03	9/24/20	Ed H	<i>V1.16</i>	RIO-SE versions 1.16.01 to 1.16.03 do not contain any additions to the API. Document version was updated to match RIO specification.
1.16.04	8/20/25	Ed H	<i>V1.16</i>	Added WATCH GROUPING and Grouping Events to allow MBX-PRE (Source Mode) and SMZ Internal Streamer to be group followers. Added SMZ to list of products supporting RIO-SE.

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Introduction

RIO Source Edition, or **RIO/SE**, is a text-based command set that brings *third party integration support* to Russound sources attached to a Russound MCA controller. RIO/SE facilitates development of UI devices and applications that require device integration and control. This protocol provides direct communication with the Russound source device, and is used as an alternative to controlling the source through communications with the Russound MCA controller. Please refer to the **“RIO Protocol for Third Party Integrators”** document for the command set used when communicating directly with the controller.

Here is a summary of the services provided:

- Supported on MBX-PRE, SMZ and MCA-88X (Internal Streamer), XSource and DMS-3.1
- Full 2-way communication capability
- The RIO/SE Command Set is available as ASCII text via TCP/IP. Up to 128 simultaneous IP connections are supported on MBX and SMZ (64 on MCA and XStream).
- TCP/IP port
 - 9621 - MBX-PRE (Source mode) and DMS-3.1
 - 9622 - SMZ and MCA-88X (Internal Streamer) and XSource (Source mode)
- Support for asynchronous notification of activity on a per-source basis
- Support for Media Management ('Content Browsing and Selection'). This provides a compressed command set, making it easy to develop a UI device capable of mimicking the **Russound App** (iOS/Android). This presents the Menu choices and Media Lists from the Russound audio devices to a third party UI via a small set of RIO/SE Media Management (MM) events. These events include support for menu browsing in several formats including JSON.
- Support for submitting multiple commands in a single request (with multiple results in a single response)
- Easy-to-understand error responses for illegal or malformed commands

Note: Third party integration should take into consideration the limited number of simultaneous RIO/SE connections, and only maintain the connection while the control system is being used.

Conventions

This section describes conventions used throughout this document. Note that these document conventions are used for clarification only. The RIO/SE Protocol is entirely case-insensitive. To be more specific, the case of incoming commands is ignored. Outgoing responses are formatted exactly as they appear in this document, regardless of the case of the incoming command.

RIO/SE commands are specified in all uppercase. For example,

```
GET S[1].name
```

In this case, 'GET' is a RIO/SE command.

Text enclosed in '< >' provides a short description of variable text data that is part of the RIO/SE message. For example,

```
S VERSION="<version #>"
```

In this case, 'version #' is a short description that is replaced with the actual RIO/SE version number at runtime.

"<key>" refers to the RIO/SE command key string.

The key syntax is described in detail in the "RIO/SE Key String" section at the end of this document.

Command Overview

RIO/SE defines a small set of commands, providing access to many of the capabilities of the Russound Source Device. These commands are presented here in general terms, along with a description of the relevant parameters. All commands are available via IP.

VERSION – request the version of the supported RIO/SE protocol

GET – return the value of one or more system parameters

SET – modify the value of one or more system parameters

EVENT – send an event from a RIO/SE client

WATCH – register to receive asynchronous messages from the specified source

RIO/SE Protocol Syntax

This section presents each RIO/SE Command and Response in detail.

The specifics of the RIO/SE protocol are:

- RIO/SE commands are case-insensitive.
- RIO/SE commands are made up of ASCII characters except for the terminating characters.
- All RIO/SE commands must be terminated with a <CR> (0x0D hex)
- All RIO/SE responses are terminated with a <CR><LF> (0x0D 0x0A hex)

RIO/SE Source Selection

The DMS-3.1 supports up to 4 sources: 3 media streamers and one AM-FM tuner. The XSource and MBX-PRE in source mode, and the MCA-88X each have a single media streamer source.

Some RIO/SE commands must be addressed to a particular source. For these commands, the name of the command is followed immediately by a source selector of the form:

S[<srcNum>]

where the <srcNum> corresponds to the source number assignment. Source numbers for Russound Media Streamers are set in the following ways:

- For DMS-3.1, the source number is taken from the switch settings on the back of the unit. Only source numbers in the range 1 to 8 are recognized. Sources that are unassigned are not addressable by RIO/SE commands.
- The source number for XSource and MBX-PRE are configured through the device web page.
- The MCA-88X internal media streamer has a hardcoded source number of 2.

Examples of valid commands using this syntax are

GET S[2].TYPE

WATCH S[2] ON

EVENT S[2]!MMInit

RIO/SE Command Response Syntax

For RIO/SE commands that are processed successfully, a response is sent with this format:

S <optional data>

For RIO/SE commands that result in failure, a response is sent with this format:

E <error message>

For asynchronous RIO/SE responses, or 'notifications', a response is sent with this format:

N <key>="<value>"

The 'VERSION' Command

The *VERSION* command is used to request the version of the RIO/SE protocol running on the device.

VERSION Command Syntax:

```
VERSION
```

VERSION Response:

```
S VERSION="<version #>"
```

VERSION Examples

1) Request the RIO/SE Protocol version:

VERSION Command:

```
VERSION
```

VERSION Response:

```
S VERSION= "01.15.04"
```

The 'GET' Command

The *GET* command returns the value of one or more system parameters. This command is performed synchronously, returning the current value once upon request. A system parameter is addressed by a 'key' string. For more details on the 'key' string, see the section at the end of this document.

To request a single system parameter,

GET Command Syntax:

```
GET <key>
```

Successful Response:

```
S <key>="<value>"
```

To request multiple system parameters,

GET Command Syntax:

```
GET <key1>, <key2>, ... , <keyN>
```

Successful Response:

```
S <key1>="<value1>", <key2>="<value2>", ... , <keyN>="<valueN>"
```

The tables below provide a complete list of the 'key' strings supported by the *GET* command.

System GET Key Table

System GET Key Table		
Key	Description	Range
System.type	Type of system	SOURCE
System.language	System Language	ENGLISH/CHINESE/RUSSIAN
System.Support.grouping	Indicates whether or not device grouping is supported.	“TRUE” – Device grouping is supported. “FALSE” – Device grouping is not supported. Note: Error is an indication of no support for device grouping.
System.Support.alarm	Indicates whether or not the device supports an alarm.	“TRUE” – Alarm is supported. “FALSE” – Alarm is not supported. Note: Error is an indication of no alarm support.
System.Support.favoritesV2	Indicates whether or not the device supports version 2 enhanced favorite features	“TRUE” – V2 favorite features are supported. “FALSE” – V2 favorite features are supported. not supported. Note: Error is an indication of no V2 favorite features support in RIO-SE. V2 features include - source, sourceType, providerMode, albumCoverURL, playerData
Notes:	System.Grouping is available if System.Support.grouping = “TRUE”	
System.Grouping.deviceList	List of devices on the network capable of grouping.	Use “WATCH Grouping” for deviceList updates. Read-only string.
System.Grouping.uuid	Unique identifier for the connected source device.	Read-only string

Source GET Key Table

Source GET Key Table		
Source table indices are identified by 's', a number from 1 to 8.		
Key	Description	Range
S[s].name	Name of the specified source	24 char max
S[s].type	Type of the specified source. <i>Note: Russound Media Streamer refers to the following:</i> • DMS-3.1 Streamer (discontinued) • MCA-88X Internal Streamer • XSource (Source mode) • MBX-PRE (Source mode)	Russound Media Streamer Russound DMS-3.1 AM/FM Tuner
S[s].ipAddress	IP Address for the specified source.	IP Address (xxx.xxx.xxx.xxx)
S[s].macAddress	MAC Address for the specified source.	MAC Address (xx.xx.xx.xx.xx.xx)
S[s].channel	Channel (frequency) of the Now Playing content on the specified source.	37 char max
<i>Notes:</i>	The 'channel' key is valid only for DMS-3.1 AM/FM Tuner	
S[s].coverArtURL	Cover Art URL for the now playing content on the specified source.	255 char max

Source GET Key Table		
Source table indices are identified by 's', a number from 1 to 8.		
Key	Description	Range
<i>Notes:</i>	The 'coverArtURL' key is valid only for Russound Media Streamer	
S[s].channelName	Channel name of the now playing content on the specified source.	37 char max
<i>Notes:</i>	The 'channelName' key is valid only for Russound Media Streamer	
S[s].genre	Genre of the now playing content on the specified source.	37 char max
<i>Notes:</i>	The 'genre' key is currently not implemented.	
S[s].artistName	Artist of the now playing content on the specified source.	37 char max
<i>Notes:</i>	The 'artistName' key is valid only for Russound Media Streamer	
S[s].albumName	Album for the now playing content on the specified source.	37 char max
<i>Notes:</i>	The 'albumName' key is valid only for Russound Media Streamer	
S[s].playlistName	Playlist for the now playing content on the specified source.	37 char max
<i>Notes:</i>	The 'playlistName' key is valid only for Russound Media Streamer	

Source GET Key Table		
Source table indices are identified by 's', a number from 1 to 8.		
Key	Description	Range
S[s].songName	Song for the now playing content on the specified source.	37 char max
<i>Notes:</i>	The 'songName' key is valid only for Russound Media Streamer	
S[s].programServiceName	Program Service Name (PSN) of the now playing content on the specified source.	37 char max
<i>Notes:</i>	The 'programServiceName' key is valid only for DMS-3.1 AM/FM Tuner	
S[s].radioText	First line of radio text for the now playing content on the specified source.	37 char max
<i>Notes:</i>	The 'radioText' key is valid only for DMS-3.1 AM/FM Tuner	
S[s].shuffleMode	Shuffle mode for the now playing content on the specified source.	OFF/ON
<i>Notes:</i>	The 'shuffleMode' key is valid only for Russound Media Streamer	
S[s].repeatMode	Repeat mode for the now playing content on the specified source.	OFF/SINGLE/ALL
<i>Notes:</i>	The 'repeatMode' key is valid only for Russound Media Streamer	

Source GET Key Table		
Source table indices are identified by 's', a number from 1 to 8.		
Key	Description	Range
S[s].mode	Provider mode or streaming service for the now playing content on the specified source. NOTE: vTuner service is indicated by the value "Internet Radio".	Unknown AirPlay Spotify Pandora SiriusXM TuneIn Internet Radio Media Server USB Airable Radio Deezer Tidal Napster Chromecast Bluetooth
<i>Notes:</i>	The 'mode' key is valid only for Russound Media Streamer	
S[s].Support.MM.longList	Source supports MM Long List commands	TRUE/FALSE Error indicates non-support
<i>Notes:</i>	The 'Support.MM.longList' key is valid only for Russound Media Streamer	
S[s].Support.MM.restoreOption	Source supports restoreOption feature	TRUE/FALSE Error indicates non-support See "Select Menu Option" section.

Source GET Key Table		
Source table indices are identified by 's', a number from 1 to 8.		
Key	Description	Range
<i>Notes:</i>	The 'Support.MM.restoreOption' key is valid only for Russound Media Streamer	
S[s].Support.availableControls	Source supports availableControls feature	TRUE/FALSE Error indicates non-support
<i>Notes:</i>	The 'Support.availableControls' key is valid only for Russound Media Streamer	
S[s].Support.trackTime	Source supports trackTime feature	TRUE/FALSE Error indicates non-support
<i>Notes:</i>	The 'Support.trackTime' key is valid only for Russound Media Streamer	
S[s].Support.playTime	Source supports playTime feature	TRUE/FALSE Error indicates non-support
<i>Notes:</i>	The 'Support.playTime key is valid only for Russound Media Streamer	
S[s].Support.playerData	Source supports playerData feature	TRUE/FALSE Error indicates non-support
<i>Notes:</i>	The 'Support.playerData key is valid only for Russound Media Streamer	

Source GET Key Table		
Source table indices are identified by 's', a number from 1 to 8.		
Key	Description	Range
S[s].Support.favoritesV2	Source supports version 2 enhanced favorites features.	TRUE/FALSE Error indicates non-support
<i>Notes:</i>	The 'Support.favoritesV2 key is valid only for Russound Media Streamer	
S[s].rating	Rating for the now playing content on the specified source.	UNKNOWN/LIKE/DISLIKE
<i>Notes:</i>	The 'rating' key is valid only for Russound Media Streamer	
S[s].playStatus	Current play status	playing paused stopped transitioning
<i>Notes:</i>		
S[s].availableControls	Playback controls that are available for the current play mode and playback state.	JSON string
<i>Notes:</i>		
S[s].sampleRate	Sample rate of the currently playing item	0 - 192000
<i>Notes:</i>		
S[s].bitRate	Sample rate of the currently playing item	0 - 999999
<i>Notes:</i>		

Source GET Key Table		
Source table indices are identified by 's', a number from 1 to 8.		
Key	Description	Range
S[s].bitDepth	Sample rate of the currently playing item	0 - 32
<i>Notes:</i>		
S[s].playTime	Play time into the current track in seconds.	0 - n
<i>Notes:</i>		
S[s].trackTime	Total track time in seconds.	0 - n
<i>Notes:</i>		

Source GET Key Table (Banks and Presets)

Source GET Key Table (Banks and Presets)		
Source table indices are identified by 's', a number from 1 to 8. Bank 'b' is a number from 1 to 6 Preset 'p' is a number from 1 to 6 Banks and Preset are valid for AM/FM Tuner types only.		
Key	Description	Range
S[s].B[b].name	Name of bank	Max 12 chars
S[s].B[b].P[p].name	Name of preset	Max 12 chars
S[s].B[b].P[p].valid	Indicate whether preset is in use	TRUE / FALSE

GET Examples

Get the IP Address of source 1:

GET Command:

GET S[1].ipAddress

GET Response:

S S[1].ipAddress="192.168.1.10"

The 'SET' Command

The *SET* command changes one or more system parameters. A system parameter is addressed by a 'key' string. For more details on the 'key' string, see the section at the end of this document.

SET operations are not subject to the current state of the system. They may be utilized at any time, provided the specified source is present in the system.

To modify a single system parameter,

SET Command Syntax:

```
SET <key>="<value>"
```

Successful Response:

```
S <key>="<value>"
```

To modify multiple system parameters,

SET Command Syntax:

```
SET <key1>="<value1>", <key2>="<value2>", ..., <keyN>="<valueN>"
```

Successful Response:

```
S <key1>="<value1>", <key2>="<value2>", ..., <keyN>="<valueN>"
```

A successful response returns the modified value.

The tables below provide a complete list of the 'key' strings supported by the *SET* command.

System SET Key Table		
Key	Description	Range
System.language	System Language	ENGLISH/CHINESE/RUSSIAN

Source SET Key Table		
Source tables indices are identified by 's', a number from 1 to 8.		
Bank 'b' is a number from 1 to 6		
Preset 'p' is a number from 1 to 6		
Banks and Preset are valid for AM/FM Tuner types only.		
Key	Description	Range
S[s].B[b].name	Bank name	Max 12 chars Banks are numbered 1..6
Notes:	See EVENT Table for setting tuner presets	

The 'EVENT' Command

The *EVENT* command is typically used to issue commands that are triggered by user actions (i.e., button presses, screen selections, etc). These commands may change system parameter values. Unlike the SET command, the EVENT command may also affect system state, such as mute status, depending on current conditions.

Due to their stateful behavior, executing the same EVENT command can provide different resultant values.

Events are directed at a source and specified by an *Event ID* and one or two event-specific data values.

EVENT Command Syntax:

```
EVENT S[s]!<event id> <data1> <data2>
```

Successful Response:

S

EVENT Table

The Events table below provides a complete list of the id's supported by the EVENT command.

Event Table			
Description	Event ID	Data 1 Range	Data 2 Range
Send a Key Release	KeyRelease	<key code> (see table below)	N/A
Send a Key Hold	KeyHold	<key code> (see table below)	hold time (in msec)
Send a UEI Key Code	KeyCode	<1-100> (see Table of Numeric Key Codes)	N/A
Change Shuffle Mode	Shuffle	N/A	N/A
Change Repeat Mode	Repeat	N/A	N/A
Set Seek Time	SetSeekTime	seconds into track	N/A
Preset Related Events (For use with Russound ST-1, ST-2, MCA-C5 internal tuner and DMS-3.1 internal tuner)			
Save a Tuner preset	SavePreset	<"preset-name"> (must be quoted)	preset number (1 to 36)
Restore a Tuner preset	RestorePreset	preset number (1 to 36)	N/A
Delete a Tuner preset	DeletePreset	preset number (1 to 36)	N/A
Set Seek Time	SetSeekTime	seconds into track	N/A
Grouping Events (See WATCH GROUPING for more details)			
Follow a device with the following uuid	Group	"<uuid>"	Group with a device with the following uuid
Detach from group	Ungroup	N/A	N/A

Key Events

Key Events are intended to emulate all of the buttons available on the Russound System Remote Control. They provide the Press/Release/Hold conditions, where applicable. The following tables list the EVENT Key Codes for each of these conditions.

Note that for Media Management source types, certain Key Codes – generally those that manage ‘transport’ functions such as ‘Play’, ‘Pause’, ‘Next’, etc. – are only processed after the source has been properly initialized. Please refer to the **Media Management** section for details.

KeyRelease Events

The table below lists the key codes that are supported by the **KeyRelease** EVENT.

RIO/SE Key Codes		
DigitZero	Previous	Enter
DigitOne	Next	Last
DigitTwo	ChannelUp	PageUp
DigitThree	ChannelDown	PageDown
DigitFour	Stop	Mute
DigitFive	Pause	
DigitSix	Play	
DigitSeven		
DigitEight		
DigitNine		

EVENT Example

Pause playback of Source 1:

EVENT Command:

EVENT S[1]!KeyRelease Pause

EVENT Response:

S

KeyHold Events

The table below lists the key codes that are supported by the **KeyHold** EVENT.

KeyHold events must be accompanied by a 'hold time' parameter. The hold time is specified in milliseconds.

In order for the KeyHold EVENTS to operate correctly, they must be executed in a specific manner by the 3rd party device (the RIO/SE 'client'). The KeyHold EVENT message must be transmitted once every 150 milliseconds for as long as the button is held. The 'hold time' parameter should be increased by 150 each time it is retransmitted. When the button is released, a KeyRelease EVENT command must be sent to complete the hold operation.

RIO/SE Key Codes		
DigitZero	Previous	Enter
DigitOne	Next	Last
DigitTwo	ChannelUp	PageUp
DigitThree	ChannelDown	PageDown
DigitFour	Stop	Mute
DigitFive	Pause	
DigitSix	Play	
DigitSeven		
DigitEight		
DigitNine		

AM/FM Tuner Specific Key Events

tunerBandToggle	"KeyRelease Play"
tunerMuteToggle	"KeyRelease Pause"
tunerUp	"KeyRelease ChannelUp"
tunerDown	"KeyRelease ChannelDown"
tunerPresetNext	"KeyRelease Next"
tunerPresetPrev	"KeyRelease Previous"
tunerBankNext	"KeyRelease PageUp"
tunerBankPrev	"KeyRelease PageDown"
tunerSeek	"KeyCode 27"
tunerScan	"KeyCode 70"
tunerStop	"KeyCode 58"
tunerStereo	"KeyCode 51"
tunerMono	"KeyCode 52"
tunerLocal	"KeyCode 66"
tunerDistant	"KeyCode 67"

Managing Banks and Presets

RIO/SE supports management of banks and presets. This is done through a combination of events as shown in the table listed in the EVENTS section and GET operations as listed in the tables in the 'GET' section.

Banks and presets operate on a per-source basis, and are only present for source types that support them, which are typically tuners.

Presets are organized into 6 banks of 6 presets each.

Bank name, Preset names and validities are accessed directly through the source with the GET command:


```
SET S[1].B[1].name="MyBank1"
```

```
S S[1].B[1].name="MyBank1"
```

```
GET S[1].B[1].name
```

```
S S[1].B[1].name="MyBank1"
```

```
GET S[1].B[1].P[2].valid
```

```
S S[1].B[1].P[2].valid="FALSE"
```

```
GET S[1].B[1].P[2].name
```

```
S S[1].B[1].P[2].name="Preset 2"
```

Note that only the Bank Name is settable through the SET command – banks are always enabled. Individual presets are managed with the EVENT command.

When managing preset with events, it is necessary to convert the Bank and Preset numbers to a single value between 1 and 36 as follows:

Bank/Preset	Numerical value
Bank 1, Preset 1	1
Bank 1, Preset 6	6
...	
Bank 2, Preset 1	7
Bank 2, Preset 6	12
...	
Bank 6, Preset 1	31
Bank 6, Preset 6	36

To save (enable) a preset with the default name, which is the station frequency, do:

```
EVENT S[1]!savePreset 1
S
GET S[1].B[1].P[1].name
S S[1].B[1].P[1].name="89.1 MHz FM"
```

To save with a user-defined name:

```
EVENT S[1]!savePreset "P1" 1
S
GET s[1].B[1].P[1].name
S S[1].B[1].P[1].name="P1"
```

Restoring and deleting presets is straightforward using the events from the Event table.

The 'WATCH' Command

The *WATCH* command enables a device to register for and receive asynchronous notifications of system parameter changes. The *WATCH* command groups the system parameters into categories: SOURCE and SYSTEM.

The *WATCH* command is an excellent way for a UI device to remain aware of system status, allowing the device to display current information with minimal communication or overhead. Each notification message is uniquely identified with a key string. This allows the UI device a way to identify relevant data, while filtering unneeded data. These UI device decisions are sometimes made on a screen-by-screen basis, or a stateful manner.

When the command is issued with the parameter set to 'ON', it will provide a snapshot of the system parameters in the requested category. Subsequent changes will be sent to the requesting device as their values change. These asynchronous change notifications will continue until the *WATCH* command is turned 'OFF' by the user or when the *WATCH* command expires (when the *EXPIRESIN* option is specified).

The *WATCH* command allows the user to specify a *WATCH* 'duration'. This value denotes the number of minutes that a particular *WATCH* operation will remain in effect. The RIO/SE Server (the Russound Source Device) will indicate when a *WATCH* is about to expire with an 'EXPIRING' notification. It will also indicate the expiration condition with an 'EXPIRED' notification.

WATCH Command Syntax:

Start a Source *WATCH* session

```
WATCH S[s] ON
```

Start a System *WATCH* session

```
WATCH System ON
```

Successful Response:

```
S
```

```
N <key1>="<value1>"
```

...

N <keyN>="<valueN>"

Start a Source WATCH session with an expiration time

WATCH S[s] ON EXPIRESIN <time in minutes>

Start a System WATCH session with an expiration time

WATCH System ON EXPIRESIN <time in minutes>

Successful Response:

S

N <key1>="<value1>"

...

N <keyN>="<valueN>"

Notification of Expiration:

N EXPIRING=SOURCE-1

N EXPIRED=SOURCE-1

Stop a Source WATCH session

WATCH S[s] OFF

S

Stop a System WATCH session

WATCH System OFF

S

WATCH Notification Messages

The type of notification messages that result from enabling WATCH vary based on the WATCH parameter and current source type. The following sections list the notifications in these various scenarios.

WATCH System

Notifications:

N System.language="<value>"

For language, possible values are ENGLISH, CHINESE and RUSSIAN.

WATCH a Source

For all source types

N S[s].name="<value>"

N S[s].type="<value>"

Where value is one of these possible source types:

- Russound Media Streamer
- Russound DMS-3.1 AM/FM Tuner

For all Russound Media Streamers

Russound Media Streamers includes DMS-3.1, MCA-88X Internal Streamer, XStream-Series and MBX-Series.

N S[s].mode="<value>"

Possible values are "Media Server", "USB", "AirPlay", "Spotify", "Chromecast", "Bluetooth", "Pandora", "Tidal", "Napster", "Deezer", "SiriusXM", "Internet Radio", "TuneIn", "Airable Radio", "SPDIF", "Line In", "Party", "Unknown"

N S[s].playlistName="<value>"

N S[s].channelName="<value>"

N S[s].artistName="<value>"

N S[s].albumName="<value>"

N S[s].songName="<value>"

N S[s].coverArtURL="<value>"

N S[s].channelArtURL="<value>"

Media Streamer Mode Metadata Support Table

The table below defines which metadata fields are available for each mode.

Media Streamer Mode:	Metadata Field						
	playlistName	channelName	artistName	albumName	songName	coverArtURL	channelArtURL
Media Server (UpnP / DLNA)	playlist / category / folder (1)	n/a	artist (2)	album (3)	song (4)	album art	n/a
USB	playlist/category/folder (1)	n/a	artist (2)	album (3)	song (4)	album art	n/a
AirPlay	"Airplay" (1)	n/a	artist (2)	album (3)	song (4)	album art	n/a
Spotify	playlist / category / folder (1)	n/a	artist (2)	album (3)	song (4)	album art	n/a
Chromecast	content description (1)	n/a	artist (2)	album (3)	song (4)	album art	n/a
Bluetooth	"Bluetooth" (1)	n/a	artist (2)	album (3)	song (4)	album art	n/a
Pandora	"Pandora" (1)	station	artist (2)	album (3)	song (4)	album art	n/a
TIDAL	playlist / category / folder (1)	n/a	artist (2)	album (3)	song (4)	album art	n/a
Napster	playlist / category/folder (1)	n/a	artist (2)	album (3)	song (4)	album art	n/a
Deezer	playlist / category / folder (1)	n/a	artist (2)	album (3)	song (4)	album art	n/a
SiriusXM	category / folder (1)	station (2)	artist	album (3)	song (4)	album art	station art *
Internet Radio (vTuner)	category / folder (1)	station (2)	artist (3)	n/a	song (4)	album art	station art *
TuneIn	category / folder (1)	station (2)	artist (3)	n/a	song (4)	album art	station art *
Airable Radio	category / folder (1)	station (2)	artist (3)	n/a	song (4)	album art	station art *
Line In	"Analog Input" (1)	n/a	n/a	n/a	n/a	n/a	n/a
SPDIF	"Digital Input" (1)	n/a	n/a	n/a	n/a	n/a	n/a
Party (Xstream-Series)	"Party" (1)	n/a	artist (2)	album (3)	song (4)	n/a	n/a
Unknown	category / folder (1)	n/a	artist (2)	album (3)	song (4)	album art	n/a

(1)– (4): Recommended order (usually top to bottom) in which text fields appear on user interface.

* Station art should be shown if no album art is available.

Russound Media Streamers (continued)

N S[s].shuffleMode="<value>"

Possible values are 'OFF' and 'ON'.

N S[s].repeatMode="<value>"

Possible values are 'OFF', 'SINGLE' and 'ALL'.

N S[s].rating="UNKNOWN"

Rating may also have the value LIKE or DISLIKE

N S[s].support.MM.longList="<value>"

Indicates if feature is supported. Possible values are 'TRUE', 'FALSE'. (no notification should be interpreted as FALSE.)

N S[s].Support.MM.restoreOption="<value>"

Indicates if feature is supported. Possible values are 'TRUE', 'FALSE'. (no notification should be interpreted as FALSE.)

N S[s].Support.availableControls="<value>"

Indicates if feature is supported. Possible values are 'TRUE', 'FALSE'. (no notification should be interpreted as FALSE.)

N S[s].Support.trackTime="<value>"

Indicates if feature is supported. Possible values are 'TRUE', 'FALSE'. (no notification should be interpreted as FALSE.)

N S[s].Support.playTime="<value>"

Indicates if feature is supported. Possible values are 'TRUE', 'FALSE'. (no notification should be interpreted as FALSE.)

N S[s].Support.playerData="<value>"

Indicates if feature is supported. Possible values are 'TRUE', 'FALSE'. (no notification should be interpreted as FALSE.)

N S[s].Support.favoritesV2="<value>"

Indicates if feature is supported. Possible values are 'TRUE', 'FALSE'. (no notification should be interpreted as FALSE.)

For Russound MBX Media Streamers

N S[s].trackTime="<value>"

N S[s].playTime="<value>"

N S[s].format="<value>"

N S[s].bitRate="<value>"

N S[s].bitDepth="<value>"

N S[s].playStatus="<value>"

For playStatus, valid values are "playing", "paused", "stopped" and "transitioning"

N S[s].availableControls="<value>"

For availableControls, values are in JSON format

Example

```
N S[1].availableControls="{\"contextMenu\":false,\"dislike\":false,\"like\":false,\"next\":false,\"pause\":false,
\"play\":true,\"previous\":false,\"repeat\":false,\"seekBack\":false,\"seekForward\":false,\"seekTime\":false,
\"seekTrack\":false,\"shuffle\":false,\"stop\":true}"
```

For Russound AM/FM Tuner Sources

N S[s].channel="<value>"

N S[s].programServiceName="<value>"

N S[s].radioText="<value>"

N S[s].channel="<value>"

WATCH Examples

1) WATCH for asynchronous changes on Source 2:

WATCH Command:

WATCH S[2] ON

WATCH Response:

S

N S[2].artistName="The Beatles"

N S[2].albumName="Abbey Road"

N S[2].songName="Come Together"

WATCH GROUPING

WATCH Command

WATCH GROUPING ON

WATCH Response

S

Notifications for MBX-PRE:

N System.Grouping.deviceList="<value>"

N System.Grouping.uuid="<value>"

Notifications for SMZ Internal Streamer:

N System.Grouping.mcaZoneList="<value>"

N System.Grouping.deviceList="<value>"

N System.Grouping.uuid="<value>"

DeviceList

System.Grouping.deviceList is a JSON formatted string. The table below lists the key-value pairs:

System.Grouping.deviceList			
Key	Description	Value	
groupId	Group identifier	Not Grouped	<i>null</i>
		Group Leader	<uuid>
		Group Follower	<uuid _{LEADER} >
ipAddress	Device IP address	String (xxx.xxx.xxx.xxx)	
name	Device name	String	
uuid	Device unique identifier	String	
volume	Device volume	String "0" - "50"	
playStatus	Play status	"stopped" "paused" "playing"	
isGroupable	Grouping is allowed	true false (Boolean)	
groupVolume	Group volume: Volume changes affect the whole group	Not Grouped	<i>null</i>
		Group Leader	String "0" - "50"
		Group Follower	<i>null</i>

mode	The streamer mode	“zone” “source”
srcNum	The source number if “source”	String “1” - “16”
primary	The primary zone of this source if configured	String “1” - “96”

Below is an example *deviceList* string with two streamers on the network:

```
{
  "devices": [
    {
      "groupId" : null,
      "ipAddress" : "192.168.10.119",
      "name" : "Office MBX",
      "uuid" : "russound-musicbox-0001",
      "volume" : "28",
      "playStatus" : "stopped",
      "isGroupable" : true,
      "groupVolume" : null,
      "mode" : "zone"
    },
    {
      "groupId" : null,
      "ipAddress" : "192.168.10.120",
      "name" : "Streamer 1",
      "uuid" : "russound-musicbox-0002",
      "volume" : "22",
      "playStatus" : "playing",
      "isGroupable" : true,
      "groupVolume" : null,
      "mode" : "source",
      "srcNum" : "1",
      "primary" : "1"
    }
  ]
}
```

Media Management

RIO/SE includes a set of commands specifically for Media Management. This allows for content filtering, browsing and selection. These commands provide access to media contained on all Russound Media Streamer source devices.

RIO/SE supports a Media Management model that allows a simple implementation to mimic the existing navigation model used on the **Russound App** (iOS/Android). By implementing four screen templates, each containing a small set of fixed buttons and static text fields, it is possible to provide full Media Management of all Russound Media Streamers. These screen templates are presented in the subsequent sections in detail. Note that Media Management is referred to as 'MM' in these sections.

RIO/SE allows for one Media Management session per IP socket connection.

Russound Media Management is supported only on Russound Media Streamers.

RIO/SE MM Commands and Responses

This section presents the RIO/SE commands and responses used to access Media Management functionality.

RIO/SE MM Commands

MM Commands are sent by the RIO/SE client application to Russound system that support RIO/SE. The first 4 RIO/SE MM commands (MMVerbosity, MMIndex, MMMaxItems and MMFormat) are intended to configure the MM session. These commands affect the way the MM responses, or 'notifications', are presented. For this reason, it is best for a RIO/SE client application to issue these commands first.

Specify Menu Notification Verbosity

This command is used to specify the amount of detail that should be provided in each MMMenuItem notification.

EVENT S[s]!MMVerbosity [1 | 2]

Verbosity level 1 results in only the MMMenuItem.text notifications to be transmitted.

Verbosity level 2 results in all verbosity level 1 notifications as well as MMMenuItem.attr notifications.

Details of the MMMenuItem are described in subsequent sections.

Specify Menu Indexing Method

This command specifies how the MMMenuItem indices are to be interpreted.

EVENT S[s]!MMIndex “[RELATIVE | ABSOLUTE]”

MMIndex RELATIVE interprets the index as a relative index of 1 thru 5 per pagination (pagination is provided via the MMPrevItems and MMNextItems commands, described later in this section). This is good for simple fixed line displays.

MMIndex ABSOLUTE interprets the indices as an absolute index of 1 to ‘max items’. This is good for touch screen or app based user interfaces which utilize smooth scrolling.

Specify Menu Notification Max Items

This command is used to specify the maximum number of items that should be delivered in response to an MM Menu operation (MMInit, MMSelectItem, MMPrevScreen, MMPrevItems and MMNextItems). This setting is only relevant when MMIndex is set to ABSOLUTE. A setting of MMIndex as RELATIVE always results in 5 items returned per request.

EVENT S[s]!MMMaxItems <1 to 255>

Specify Menu Notification Format

This command specifies the format to be used for the MMMenuItem Notification messages. MMMenuItem RIO/SE notifications always begin with the letter ‘N’ and are sent as individual messages per menu item. JSON menu item notifications are packed into a single JSON-formatted string per each pagination command. The number of items contained within a single JSON notification is specified by using the MMMaxItems command. You can expect to get better performance from the JSON format. For Javascript-based RIO/SE client applications, notification parsing will be easier as well. See **JSON Response Format** section below.

EVENT S[s]!MMFormat “[RIO | JSON]”

JSON Response Format

If MMFormat is set to JSON the menu item notifications will be return in the following format:

Field Descriptions

totalItems - the total number of items available in this menu

numItems - the number of items in this notification list

menuItems - an array of menu items containing these fields:

id - the instance of this menu item

text - the text to be displayed for this menu item

If the MMVerbosity is set to 2, these JSON elements should be present:

isFirst: if true, this is the first menu item in the menu

isLast: if true, this is the last menu item in the menu

isMenu: if true, this menu item is a menu selection

BOT: If true, indicates that this item is the first item in a given transmission.

EOT: If true, indicates that this item is the last item in a given transmission.

value: (optional) The value of the item to be displayed (not to be confused with 'text')

imgURL: (optional) URL of associated album art or station art.

uri: (optional) URI associated with items that have Invokable or Hyperlink attributes.

Attributes: (optional) list of attributes (see Menu Item Attributes section).

Note: There are also attributes for isFirst, isLast, isMenu. If attributes are defined, isFirst, isLast, isMenu can be treated as deprecated.

Example:

```
{ "totalItems": 37, "numItems": 37, "menuItems":
[
  { "id": 1, "text": "FM", "isFirst": true, "isLast": false, "isMenu": false, "BOT": true, "EOT": false, "attributes": "Bfh" },
```



```

{"id":2,"text":"89.1 | NHPR (US News) | The Moth","imgURL":"http://cdn-radiotime-
logos.tunein.com/s28632q.png","isFirst":false,"isLast":false,"isMenu":false,"BOT":f
alse,"EOT":false,"attributes":"JcS"},
...
{"id":3,"text":"Embedded library version","value":"HEAD-v3.52.18-
g36a3fdc3","isFirst":false,"isLast":true,"isMenu":false,"BOT":false,"EOT":true,"att
ributes":"ELJsd"}

{"id":4,"text":"Open Spotify
App","uri":"spotify:/openApp","isFirst":true,"isLast":false,"isMenu":false,"BOT":tr
ue,"EOT":false,"attributes":"BFJI"}
...
] }

```

Specify Info Screen Format

This command specifies how to receive INFO screen text. If set to TRUE, the text arrives in one block. If set to FALSE the text arrives in up to 4 separated text fields.

EVENT S[s]!MMUseBlockInfo [TRUE | FALSE]

Specify Text Entry Format

This command specifies how to receive TEXT entry prompts. If set to TRUE, all text entry prompts arrive at once. If set to FALSE, text entry prompts arrive one at a time.

EVENT S[s]!MMUseForms [TRUE | FALSE]

Navigate to a Menu Item by Index

This EVENT is used to navigate to a menu item that was received via notification. Subsequent MMPrevItems and MMNextItems will be made relative to the new menu position.

EVENT S[s]!MMStartItem <1 to 2³²>

Note: This command is only available when MMIndex is set to ABSOLUTE.

MM Session Initialization

This event restarts MM menu state within the source device. The menu navigation is reset to the topmost (i.e., home) menu. Browsing menus and ultimately making a selection of a playable item will bring the session to the NowPlaying screen.

EVENT S[s]!MMInit

If it is advantageous for the session to respond to Key Events for transport controls (e.g. Play, Pause, Next, etc) *before* a playable selection has been made, the session may be moved directly to the NowPlaying screen by using the MMClose command (see below).

MM Session Termination

This event ends the MM session and returns the RIO/SE session to the NowPlaying screen.

EVENT S[s]!MMClose

After issuing this RIO/SE MM command, a source-specific NowPlaying screen notification will be sent to the RIO/SE client. The NowPlaying screen notification is described in the **Screen Change Notifications** section below.

Moving the MM session to the NowPlaying screen enables Key Events for transport controls (e.g. Play, Pause, Next, etc) to be processed.

Menu Item Selection

This EVENT is used to select from a list of items presented by the previous set of text field notifications (see **Menu Item Notifications** below). The number of valid selections is dependent on the number of MMItems that are received from the RIO/SE source device and the value of MMIndex (RELATIVE or ABSOLUTE), depending on the current screen being displayed.

If MMIndex is set to RELATIVE,

EVENT S[s]!MMSelectItem [1 to 6]

If MMIndex is set to ABSOLUTE,

EVENT S[s]!MMSelectItem [1 to max items]

Setting Optional Value

MMSelectItem can also be used to set an optional value when selected.

EVENT S[s]!MMSelectItem [1 to 6] [<value>]

If MMIndex is set to ABSOLUTE,

EVENT S[s]!MMSelectItem [1 to max items] [<value>]

Setting values is only valid when used with menu items that have the following attributes:

Editable

EditTypeString

EditTypeIPAddress

EditTypeNumber

EditTypeSlider

EditTypeRadioBox

EditTypeBool

Item Context Menu Selection

This EVENT is used to select the context menu of an item in a list of items presented by the previous set of text field notifications (see **Menu Item Notifications** below). The number of valid selections is dependent on the number of MMItems that are received from the RIO source device and the value of MMIndex (RELATIVE or ABSOLUTE), depending on the current screen being displayed. This command should only be used if the menu item attribute includes the HasContextMenu attribute.

If MMIndex is set to RELATIVE,

EVENT S[s]! MMItemContextMenu [1 to 6]

If MMIndex is set to ABSOLUTE,

EVENT S[s]! MMItemContextMenu [1 to max items]

Screen Data Confirmation

This EVENT is used as confirmation for the current screen data and advances the Media Management session to the next navigable screen. The Info and TextEntry screens for the Russound Media Streamer make use of this MM EVENT. These screens are described in more details in subsequent sections.

EVENT S[s]!MMSelectOk

Item Navigation

These EVENTS are used to request the next and previous set of items from the current list.

EVENT S[s]!MMNextItems

EVENT S[s]!MMPrevItems

Screen Navigation

This EVENT requests that the previous screen be selected. This may result in the system sending a screen change notification (see **Screen Change Notifications** below) as well as text field change notifications (see **Menu Item Notifications** below).

EVENT S[s]!MMPrevScreen

Text Edit Navigation

These EVENTS allow editing of alphanumeric fields, where the text search feature is available.

EVENT S[s]!MMCursorNext

EVENT S[s]!MMCursorPrev

EVENT S[s]!MMLetterUp

EVENT S[s]!MMLetterDown

The Context Menu

Once a selection has been made, the user interface will be brought to the Now Playing screen which displays information about the current selection. Certain services, such as Pandora, support retrieval of additional context specific information. A Context Menu is invoked by using:

EVENT S[s]! MMContextMenu

This event invokes a new set of menu items containing choices relative to the particular service. Navigation of the context menus is done using the same events as are used for the other selection menus.

Likes and Dislikes

Some services support the ability for the user to express preferences about the current selection. This is done with an event:

EVENT S[s]!MMRate high | low

Where *high* is LIKE and *low* is DISLIKE.

When songs change, rating information is sent as notifications that appear when the source is being watched. They will appear as:

N S[2].rating="UNKNOWN"

This indicates no rating information for that song has been previously provided, and is the default behavior. If a preference has been provided, values of “LIKE” or “DISLIKE” will be shown.

Text Field Entry

This EVENT allows direct entry of a text field, where the text search feature is available. This approach should be used by advanced UI devices capable of presenting an ASCII keyboard to the user for text entry. The TextEntry screen for the Russound Media Streamer makes use of this MM EVENT.

EVENT S[s]!MMTextField "<text string>"

Select Menu Option

Used in top level service selection. Music service attributes not containing ‘M’ (sub-menu item) indicate that the previously played station, channel or song for that service will be played upon selection.

“default” – Default behavior as explained above.

“norestore” – This event option allows you to override the content restore and enter the sub-menu for the selected music service.

EVENT S[s]!MMSelectOption “norestore”

Keep Alive

Call this to keep media management from timing out (1 minute) due to lack of user input.

EVENT S[s]!MMKeepAlive

RIO/SE MM Responses

Screen Change Notifications

This section presents the RIO/SE messages **received by** the Third Party Device to indicate that a new screen template must be displayed. This RIO/SE message is typically followed by a series of RIO/SE messages containing text, used to populate the text fields within the screen to be displayed. For more information on these text fields messages, see the **Menu Item Notifications** section.

N S[s].MMScreen="<Screen ID>"

where Screen ID is defined as:

Russound Media Streamer Screen Identifiers

Screen ID	Screen Name
SourceMenuScreen	Menu
SourceInfoScreen	Info
SourceNowPlayingScreen	Now Playing
SourceTextEntryScreen	Text Entry

Screen Title Notifications

All screens can expect notifications of changes to the screen title. These will appear as notification messages in this format:

Screen Title Text

N S[s].MMTitle.text="<title string>"

Screen Title Attributes

If MMVerbosity is set to '2', title attribute notifications will also be transmitted. Valid title attributes include HomeMenu, ("H"), UserLogin ("U"), any one of the streaming service attributes. Combination of UserLogin and a streaming service attribute identify the screen as the login screen for a specific service (i.e. SiriusXM).

N S[s].MMTitle.attr=<text string>

Menu Item Notifications

Menu screens can expect notifications of changes to Menu Item text. These will appear as notification messages in this format:

N S[s].MMMenuItem[1 to 2^32].text="<text string>"

Menu Item Text

N S[s].MMMenuItem[1 to 2^32].text="<text string>"

Menu Item Attributes

If MMVerbosity is set to '2', menu item attribute notifications will also be transmitted:

N S[s].MMMenuItem[1 to 2^32].attr="<text string>"

The attr string is a series of letters that provide further details about the menu item. Note that a text string may have multiple attributes. The following table lists the display attributes used in MenuItem notifications.

Attribute	Attribute Key	Definition	New For MBX
BOT	“B”	Beginning Of Transmission. Marker for first in a list of indeterminate length.	
EOT	“E”	End of Transmission. Marker for last in a list of indeterminate length.	
LastMenuItem	“L”	Indicates this is the last item in a menu. Useful for hiding/showing the Page Down control.	
SubMenuItem	“M”	Sub-menu item.	
FirstMenuItem	“F”	Indicates this is the first item in a menu. Useful for 'hiding' the Page Up Control	
Hidden	“X”	Menu item is hidden	
CurrentMenuItem	“C”	Indicates the currently selected menu item. This should only be asserted to indicate the item 'uniquely' on the UI.	
ActiveMenuItem	“A”	Indicates the menu item is in an active state. Usually to indicate a setting state.	
PlayingMenuItem	“P”	Indicates the current menu item is playing	
HomeMenu	“H”	Indicates a home or top menu. Typically used when transmitting the menu title text.	
UserLogin	“U”	Indicates that the screen is a login screen. This is normally paired with one of the service type attributes below to identify a specific service login screen. Typically used when transmitting the menu title text	
JSON	“J”	Display message in JSON format	

Subtext	“S”	Display message has subtext delimited by “ “. *
Invokable	“I”	Item is invokable and should be displayed as a button *
Header	“h”	Item is a non actionable menu header. *
HasContextMenu	“c”	Item has a context menu *
Hyperlink	“w”	Item has a hyperlink *
Editable	“e”	Item is editable *
EditTypeString	“s”	Edit type is string *
EditTypeIPAddress	“i”	Edit type is IP address *
EditTypeNumber	“n”	Edit type is number *
EditTypeSlider	“I”	Edit type is slider *
EditTypeExecute	“x”	Edit type is executable (typically shown as button or link text) *
EditTypeRadioBox	“r”	Edit type is radio box control *
EditTypeBool	“b”	Edit type is bool (typically shown as switch or check box) *
Disabled	“d”	Item is disabled *
Pandora	“!”	Service topmost menu item or service specific screen title.
TuneIn	“@”	Service topmost menu item or service specific screen title.
SiriusXM	“#”	Service topmost menu item or service specific screen title.
vTuner	“\$”	Service topmost menu item or service specific screen title.
DLNA	“%”	Service topmost menu item or service specific screen title.
USB	“^”	Service topmost menu item or service specific screen title.
Spotify	“&”	Service topmost menu item or service specific screen title.

Party	“*“	Service topmost menu item or service specific screen title.	
Tidal	“(“	Service topmost menu item or service specific screen title.	*
Napster	)“	Service topmost menu item or service specific screen title.	*
Deezer	“-“	Service topmost menu item or service specific screen title.	*
Bluetooth	“ _ “	Service topmost menu item or service specific screen title.	*
SPDIF	“+“	Service topmost menu item or service specific screen title.	*
LineIn	“[“	Service topmost menu item or service specific screen title.	*
Chromecast	“]“	Service topmost menu item or service specific screen title.	*
AirableRadio	“ “	Service topmost menu item or service specific screen title.	*

In response to the pagination commands, ***MMPrevItems*** and ***MMNextItems***, these notifications are transmitted:

MMMenu.totalItems

This notification indicates the total number of items present in the current menu

N S[s].MMMenu.totalItems=“”

MMMenu.numItems

This notification indicates the number of items present in the most recent pagination request:

N S[s].MMMenu.numItems=“”

Info Screen Notifications

Info Screen Buttons

Info screens may need to display 0 or more buttons to indicate what actions are required for the screen. Each notification indicates the type of the button and the string to be displayed to the user for that button:

```
N S[2].MMBtnOK.text="<text string>"
```

```
N S[2].MMBtnBack.text="<text string>"
```

If no button notifications are present for a screen, no buttons should be displayed.

Info Screen Text

Info screens display 4 lines of text, which appear as text with attributes:

```
N S[2].attr="BF"
```

```
N S[2].MMInfo[1].text="If you want to logout of"
```

```
N S[2].MMInfo[2].text="SiriusXM, press OK. If not,"
```

```
N S[2].MMInfo[3].text="press the Back/Prev button."
```

```
N S[2].attr="EL"
```

```
N S[2].MMInfo[4].text=""
```

If there is not enough text to fill 4 lines, empty lines will be provided as padding. The first and last lines are indicated by attributes; if more than 4 lines are present, the text may be scrolled with KeyRelease PageUp and KeyRelease PageDown events.

Info Screen Changes

The Info screen can expect notifications of changes to the info fields. These will appear as notification messages in this format:

```
N S[s].MMInfo[1-4].text="<info string>"
```

Now Playing Notifications

Source notifications such as metadata and volume drive updates of the Now Playing screen. Source notifications may arrive prior to a screen change notification, so values should be cached as they are received.

Like and Dislike

When songs change, rating information is sent as notifications that appear when the source is being watched. They will appear as one of the following:

```
N S[2].rating="UNKNOWN"
```

```
N S[2].rating="LIKE"
```

```
N S[2].rating="DISLIKE"
```

Text Entry Screen Notifications

The Text Entry screen can expect notifications of changes to the text entry field (where user text is entered) and help text (describing what type of information the user should enter into the text entry field). These will appear as notification messages in this format:

```
N S[s].MMTextField.text="<text string>"
```

```
N S[s].MMHelp.text="<text string>"
```

Forms

It is possible to have multiple prompts and multiple fields on a single Text Entry screen for user input. This is referred to as “using forms”. Using forms is enabled with an event:

```
EVENT S[1]!MMUseForms true
```

Once this is done, if the Russound Media Streamer has text entry screens with multiple input lines (e.g. a screen with ‘username’ and ‘password’) the following notifications will be seen:

```
N S[2].MMScreen="SourceTextEntryScreen"
```

```
N S[2].MMTitle.text="SiriusXM"
```

```
N S[2].MMHelp.text="Enter your Username"
```

```
N S[2].MMForm1Field.text="Russound40"
```

```
N S[2].MMHelp2.text="Enter your Password"
```

```
N S[2].MMForm2Field.text="siriusxm"
```

Note that the MMHelp field is augmented by field ‘MMHelp2’. The MMTextField is replaced by ‘MMForm1Field’ and ‘MMForm2Field’.

When MMUseForms is true, you may set the text fields in any order (and as often as you like) using the MMFormEntry event:

```
EVENT S[1]!MMFormEntry "user" 1
```

```
EVENT S[1]!MMFormEntry "password" 2
```

Where the first argument is the string to be used and the second argument is the number of the MMForm<n>Field to which the string is to be applied.

Once all Form Fields have been set, use

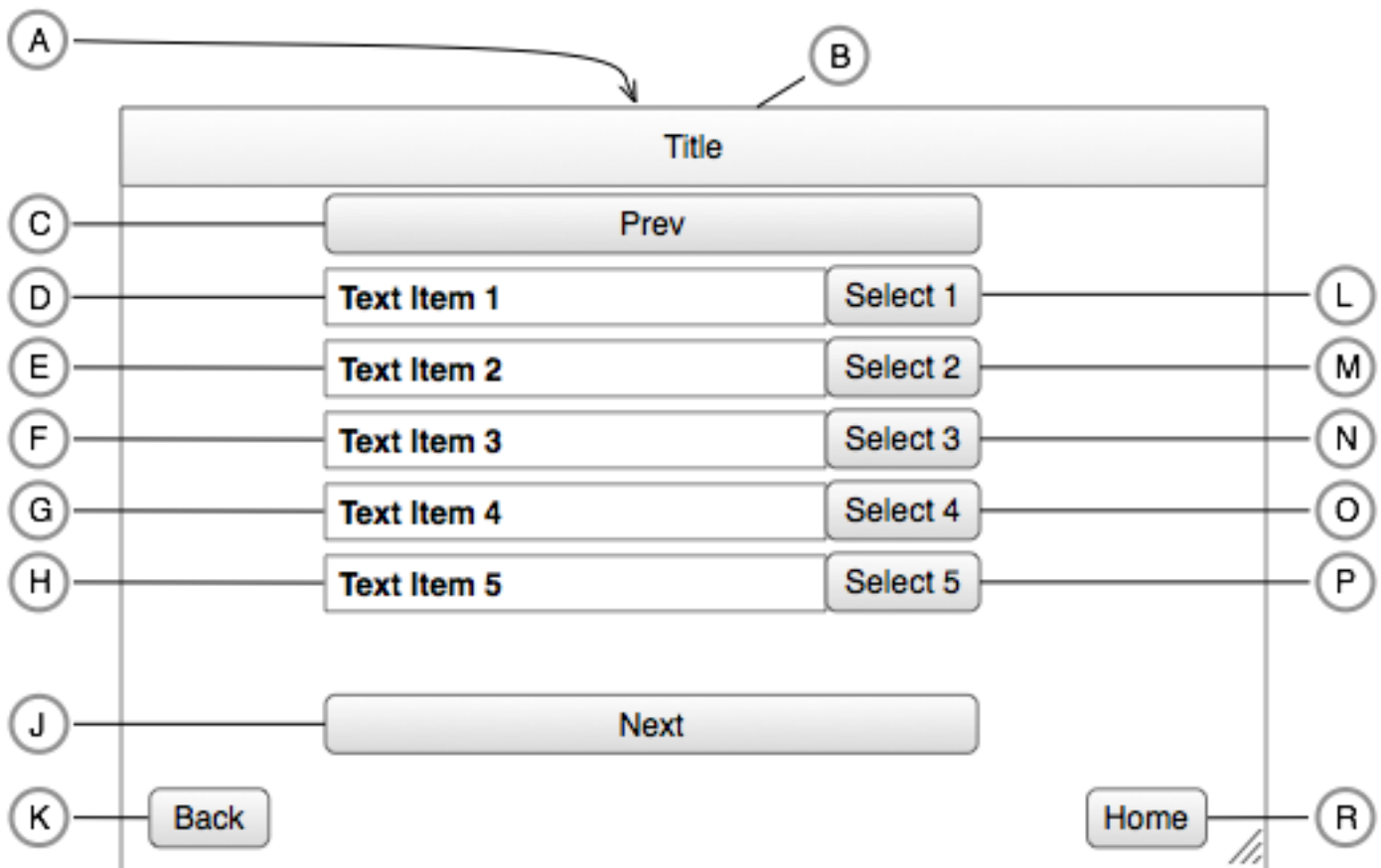
EVENT S[1]!MMSelectOk

to submit the form and move on to subsequent screens

Russound Media Streamer Example Screens

Menu Screen

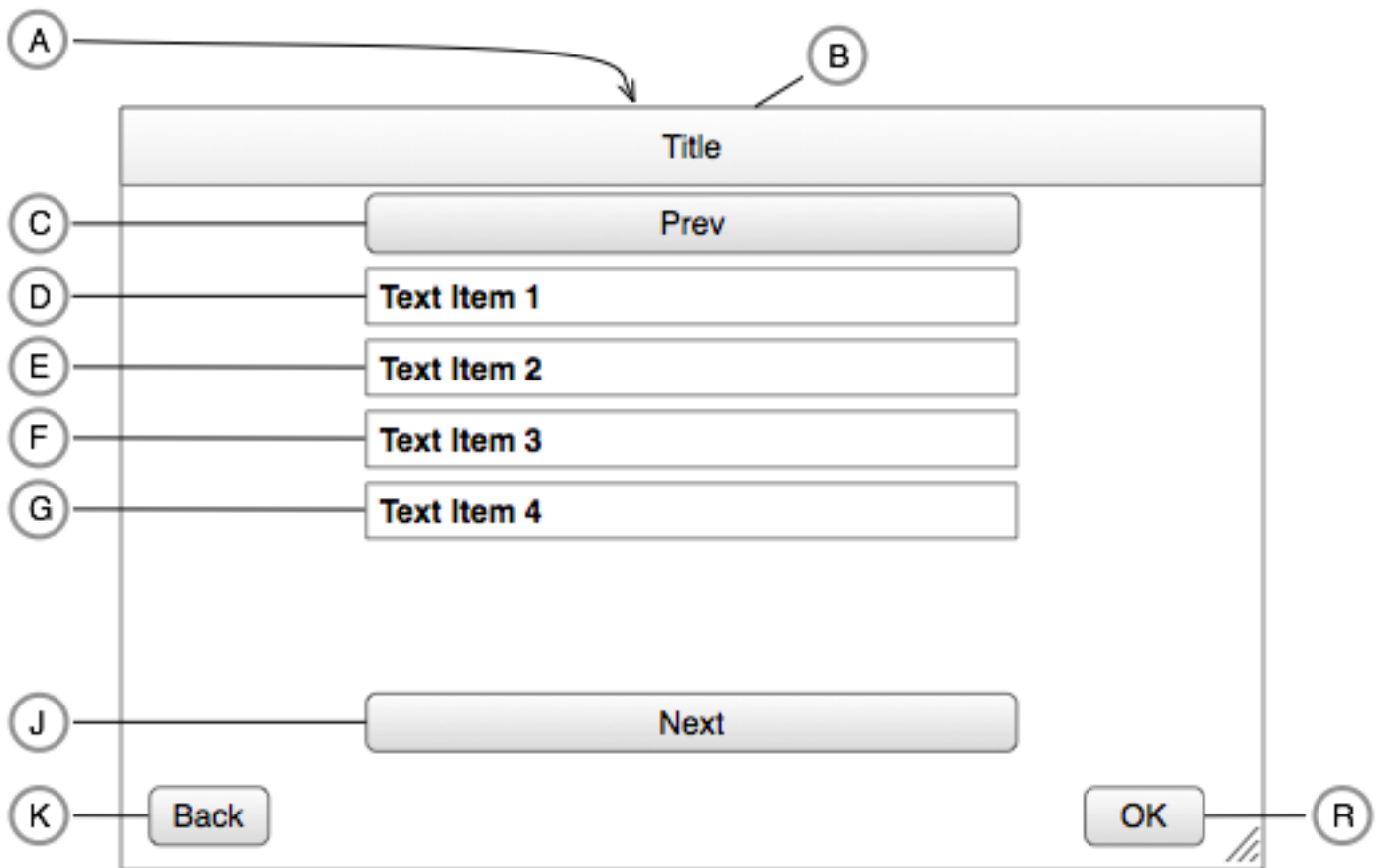
The example below uses MMIndex set to RELATIVE.



Menu Screen Elements

Screen Designator	RIO/SE Command/Response
A	N S[s].MMScreen="SourceMenuScreen"
B	N S[s].MMTitle.text="<title string>"
C	EVENT S[s]! KeyRelease PageUp
D	N S[s].MMMenuItem[1].text="<text string>"
E	N S[s].MMMenuItem[2].text="<text string>"
F	N S[s].MMMenuItem[3].text="<text string>"
G	N S[s].MMMenuItem[4].text="<text string>"
H	N S[s].MMMenuItem[5].text="<text string>"
I	N/A
J	EVENT S[s]!KeyRelease PageDown
K	EVENT S[s]!MMPrevScreen
L	EVENT S[s]!MMSelectItem 1
M	EVENT S[s]!MMSelectItem 2
N	EVENT S[s]!MMSelectItem 3
O	EVENT S[s]!MMSelectItem 4
P	EVENT S[s]!MMSelectItem 5
Q	N/A
R	EVENT S[s]!MMInit
S	N/A
T	N/A
U	N/A
V	N/A

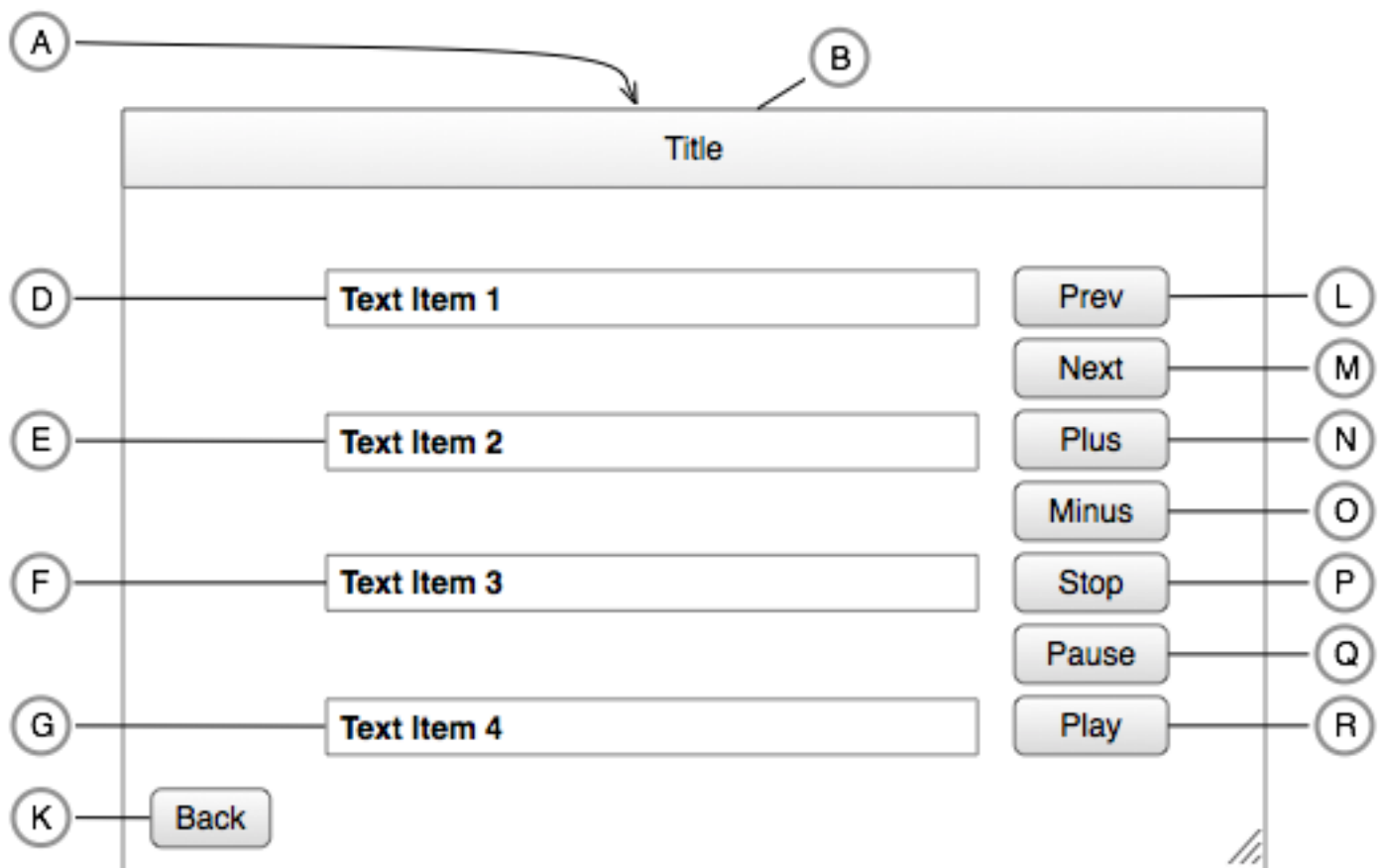
MM Info Screen



Info Screen Elements

Screen Designator	RIO/SE Command/Response
A	N S[s].MMScreen="SourceInfoScreen"
B	N S[s].MMTitle.text="<title string>"
C	EVENT S[s]!KeyRelease PageUp
D	N S[s].MMMenuItem[1].text="<text string>"
E	N S[s].MMMenuItem[2].text="<text string>"
F	N S[s].MMMenuItem[3].text="<text string>"
G	N S[s].MMMenuItem[4].text="<text string>"
H	N/A
I	N/A
J	EVENT S[s]!KeyRelease PageDown
K	EVENT S[s]!MMPrevScreen
L	N/A
M	N/A
N	N/A
O	N/A
P	N/A
Q	N/A
R	EVENT S[s]!MMSelectOk
S	N/A
T	N/A
U	N/A
V	N/A

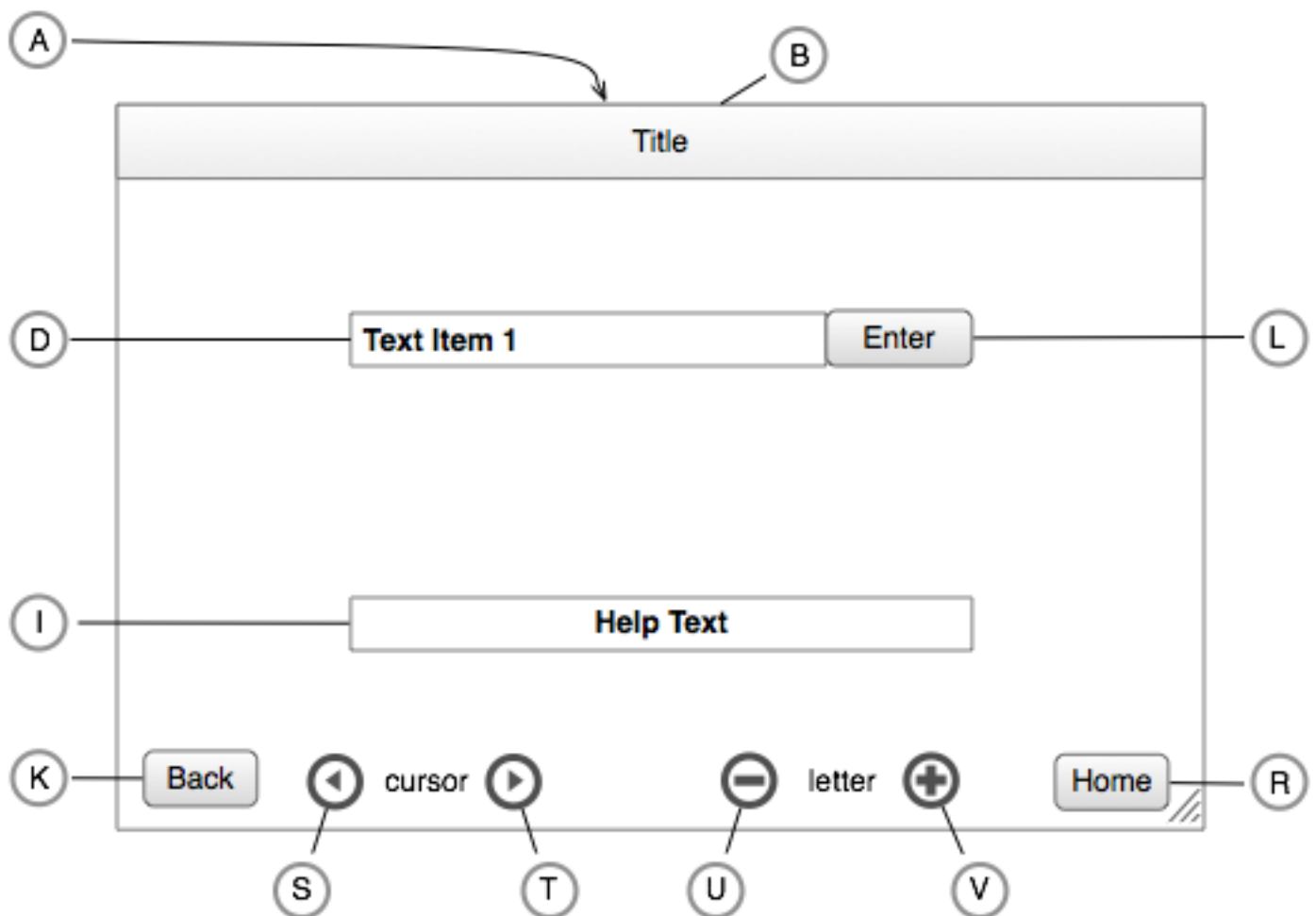
Now Playing Screen



Now Playing Screen Elements

Screen Designator	RIO/SE Command/Response
A	N S[s].MMScreen="SourceNowPlayingScreen"
B	N S[s].MMTitle.text="<title string>"
C	N/A
D	N S[s].playlistName="<value>" / N S[s].channelName="<value>"
E	N S[s].artistName="<value>"
F	N S[s].albumName="<value>"
G	N S[s].songName="<value>"
H	N/A
I	N/A
J	N/A
K	EVENT S[s]!MMPrevScreen
L	EVENT S[s]!KeyRelease Previous
M	EVENT S[s]!KeyRelease Next
N	EVENT S[s]!KeyRelease ChannelUp
O	EVENT S[s]!KeyRelease ChannelDown
P	EVENT S[s]!KeyRelease Stop
Q	EVENT S[s]!KeyRelease Pause
R	EVENT S[s]!KeyRelease Play
S	N/A
T	N/A
U	N/A
V	N/A

Text Entry Screen



Text Entry Screen Elements

Screen Designator	RIO/SE Command/Response
A	N S[s].MMScreen="SourceTextEntryScreen"
B	N S[s].MMTitle.text="<title string>"
C	N/A
D	N S[s].MMTextField.text="<text string>"
E	N/A
F	N/A
G	N/A
H	N/A
I	N S[s].MMHelp.text="<text string>"
J	N/A
K	EVENT S[s]!MMPrevScreen
L	EVENT S[s]!MMTextField "<text string>"
M	N/A
N	N/A
O	N/A
P	N/A
Q	N/A
R	N/A
S	EVENT S[s]!MMCursPrev
T	EVENT S[s]!MMCursNext
U	EVENT S[s]!MMLetterUp
V	EVENT S[s]!MMLetterDown

Summary of RIO/SE Events

KeyRelease
KeyHold
KeyCode
Shuffle
Repeat
SavePreset
RestorePreset
DeletePreset

Summary of RIO/SE Media Management Events

MMInit	MMSelectOk
MMClose	MMTextField
MMSelectItem	MMVerbosity
MMNextItems	MMFormat
MMPrevItems	MMIndex
MMPrevScreen	MMRate
MMCursorNext	MMUseForms
MMCursorPrev	MMUseBlockInfo
MMLetterUp	MMFormEntry
MMLetterDown	MMSelectOption
MMMaxItems	MMContextMenu
MMStartItem	MMKeepAlive
MMItemContextMenu	
<i>DEPRECATED</i> <i>MMStartChar – use MMStartItem</i>	

Summary of Media Management INFO Screen Notifications

The following INFO screen notifications pertain to DMS-3.1 and XStream-Series.

Pandora Acct Inactive Info

Title: "Inactive Account"

Text: "Sorry, this account is no longer active. Please contact Pandora listener support at pandora-support@pandora.com."

BtnOK: "OK"

Pandora Confirm Logout

Title: "Sign Out"

Text: "Are you sure that you want to Sign Out of Pandora?"

BtnOK: "OK"

BtnBack: "Cancel"

Pandora Empty Search Info

Title: "Error"

Text: "You entered a blank string. Please try again."

BtnOK: "OK"

Pandora Invalid Country

Text: "Sorry, Pandora is not available in this country"

BtnOK: "OK"

Pandora Invalid Login

Title: "Login Failed"

ONE OF THE FOLLOWING:

Text: "User must sign in to re-authenticate the device."

Text: "Username or password is invalid."

Text: "Sorry, this account is no longer active. Please contact Pandora listener support at pandora-support@pandora.com."

Text: "User is not authorized."

Text: "Pandora is performing system maintenance. Please try again later."

Text: "Complimentary period for this device has expired."

Text: "Unable to communicate with Server. Please check your network settings and try again."

Text: "This device is not registered with Pandora."

Text: "Device partner is not authorized for this action."

Text: "Username is invalid."

Text: "Password is invalid."

Text: "Username provided is already in use."

Text: "This device is already associated to an account."

Text: "Email address is invalid."

Text: "This device model is invalid."

Text: "User account is not found."

Text: "Internal server error."

BtnOK: "OK"

Pandora Invalid Logout

Title: "Logout Failed"

Text: "Pandora is performing system maintenance. Please try again later"

BtnOK: "OK"

Pandora Option Delete Station

Title: "Delete Station"

Text: "Are you sure that you want to delete [currentStation]?"

BtnOK: "Delete"

BtnBack: "Cancel"

Sirius Confirm Logout

Title: "Logout"

Text: "Are you sure that you want to logout of SiriusXM?"

BtnOK: "OK"

BtnBack: "Cancel"

Sirius Invalid Login

Title: "Login Failed"

ONE OF THE FOLLOWING:

Text: "General Error (SiriusXM). Please contact your Russound Installer."

Text: "Unknown Error (SiriusXM). Please contact your Russound Installer."

Text: "Invalid Device (SiriusXM). Please contact your Russound Installer."

Text: "Service Not Found (SiriusXM). Please contact your Russound Installer."

Text: "Upgrade Required (SiriusXM). Please contact your Russound Installer."

Text: "Missing Parameter Error (SiriusXM). Please contact your Russound Installer."

Text: "Service Unavailable (SiriusXM). Please contact your Russound Installer."

Text: "Firmware Update Required (SiriusXM). Please contact your Russound Installer."

Text: "Invalid Device ID (SiriusXM). Please contact your Russound Installer."

Text: "Device Temporarily Unavailable (SiriusXM). Please try again later."

Text: "Device Disabled (SiriusXM). Please contact your Russound Installer."

Text: "Authentication Required (SiriusXM). Please contact your Russound Installer."

Text: "Authentication password is incorrect (SiriusXM). Please contact your Russound Installer."

Text: "Authentication Failed (SiriusXM). Please contact your Russound Installer."

Text: "Authentication session has expired (SiriusXM). Please contact your Russound Installer."

Text: "Authentication (SiriusXM): Invalid Sequence Error. Please contact your Russound Installer."

Text: "Authentication (SiriusXM): Bad Session Error. If this persists, please contact Russound Technical Support."

Text: "Password expired (SiriusXM). Please contact your Russound Installer."

Text: "Invalid Client Certificate (SiriusXM). Please contact your Russound Installer."

Text: "Services Login Required. Please contact your Russound Installer."

Text: "The Username and Password are not valid. Please try again or contact SiriusXM to subscribe."

Text: "External Authorization Required (SiriusXM). Please contact your Russound Installer."

Text: "Channel Lineup Out Of Date (SiriusXM). Please contact your Russound Installer."

Text: "Insufficient Authentication Level Error (SiriusXM). Please contact your Russound Installer."

Text: "Radio Authentication Failure (SiriusXM). Please contact your Russound Installer."

Text: "Free SiriusXM trial expired. Please contact SiriusXM to subscribe."

Text: "Unable to Login. Please contact SiriusXM support."

Text: "Unable to Login. Please contact SiriusXM support."

Text: "Unable to Login (SiriusXM). You have exceeded the maximum daily logins."

Text: "Unknown error. Please contact your Russound Installer."

BtnOK: "OK"

TuneIn Acct Association Info

Title: "Account Association"

if (Associated)

Text: "Your account is associated with [accountName]"
 "To disassociate this device from your account, visit <http://tunein.com/devices>"

else

Text: "Your Association Code is [associationCode]. To associate this device with your account, visit <http://tunein.com/devices>"
 "To create an account, visit <http://tunein.com>"

BtnOK: "OK"

TuneIn Search Info

Title: "TuneIn Search Info"

ONE OF THE FOLLOWING:

Text: *Search Results*

Text: "No search results for [search string]. Please try again."

BtnOK: "OK"

Party Leave

Title: "Leave Party?"

Text: " Would you like to leave this party?"

BtnOK: "Leave"

BtnBack: "Cancel"

Party Share

Title: "Start Sharing?"

Text: "Would you like to start sharing?"

"You can join this party from another room by selecting Party->Join Party."

BtnOK: "Share"

BtnBack: "Cancel"

Party Share AirPlay

Title: "Could not share."

Text: "I'm sorry. Due to licensing restrictions, it is not possible to share AirPlay content."

"To start sharing, please select a different content type."

BtnOK: "OK"

Party Share Full

Title: "Could not share."

Text: "I'm sorry, there are no party sessions available."

"You can create up to two parties, with up to six rooms participating in each party at any given time."

"To create a new party, stop sharing from one of the other rooms by selecting Party->Stop Sharing."

BtnOK: "OK"

Party Share Unsupported Format

Title: "Could not share."

if (Spotify)

Text: "I'm sorry, Spotify content cannot be shared."

"If you would like to start a new party session, please select different content."

else

Text: "I'm sorry, High Definition audio cannot be shared."

"If you would like to start a new party session, please select different content."

BtnOK: "OK"

Party Stop

Title: "Could not share."

Text: "Would you like to stop sharing?"

BtnOK: "Stop"

BtnBack: "Cancel"

AirPlay In Use

Title: "AirPlay In Use"

Text: "To make another media selection, please discontinue AirPlay streaming."

BtnOK: "OK"

Pandora Option Info

ONE OF THE FOLLOWING:

Text: "Station deleted, please select a new station."

Text: "Pandora will play more music like this."

Text: "This track won't play again on this station."

Text: "Station Created."

Text: "Station Saved."

Text: "This option is not currently available."

BtnOK: "OK"

Server Error

Title: "Error"

Text: *Error Message*

BtnOK: "OK"

RIO/SE Key String

The key string is formatted as a dot-separated (‘.’) series of strings that refer to a hierarchical set of branches, tables and leafs.

Key strings are case insensitive.

Branches, represented as a capitalized string, serve to organize the system parameters by category. A key string can contain multiple branches.

For example, in the WATCH SYSTEM command response,

```
N System.status="<value>"
```

‘System’ is a branch string.

Tables, represented by a capitalized string followed by a bracketed 1-based number, allow for instances of sources to be referenced by index. A key string can contain multiple tables.

For example, in the WATCH Source command response,

```
N S[1].type="<value>"
```

‘S[s]’ is a table.

Leafs, represented by a lowercase ‘camel’ string and always the last string in the key string, refer to a specific system parameter. A key string has only one leaf string.

For example, in the WATCH Source command response,

```
N S[1].rating="LIKE"
```

‘rating’ is a leaf string.

Using PuTTY as a RIO/SE Client

PuTTY is a popular free implementation of Telnet and SSH for Win32 and Unix platforms. It is possible to use PuTTY as a RIO/SE client for testing purposes, provided it is configured correctly.

Available here: <http://www.chiark.greenend.org.uk/~sgtatham/putty/download.html>

- Enter hostname or IP address
- Set connection type to Raw
- Set Port to 9622 (9621 for MBX-PRE and DMS-3.1)
- Select Open

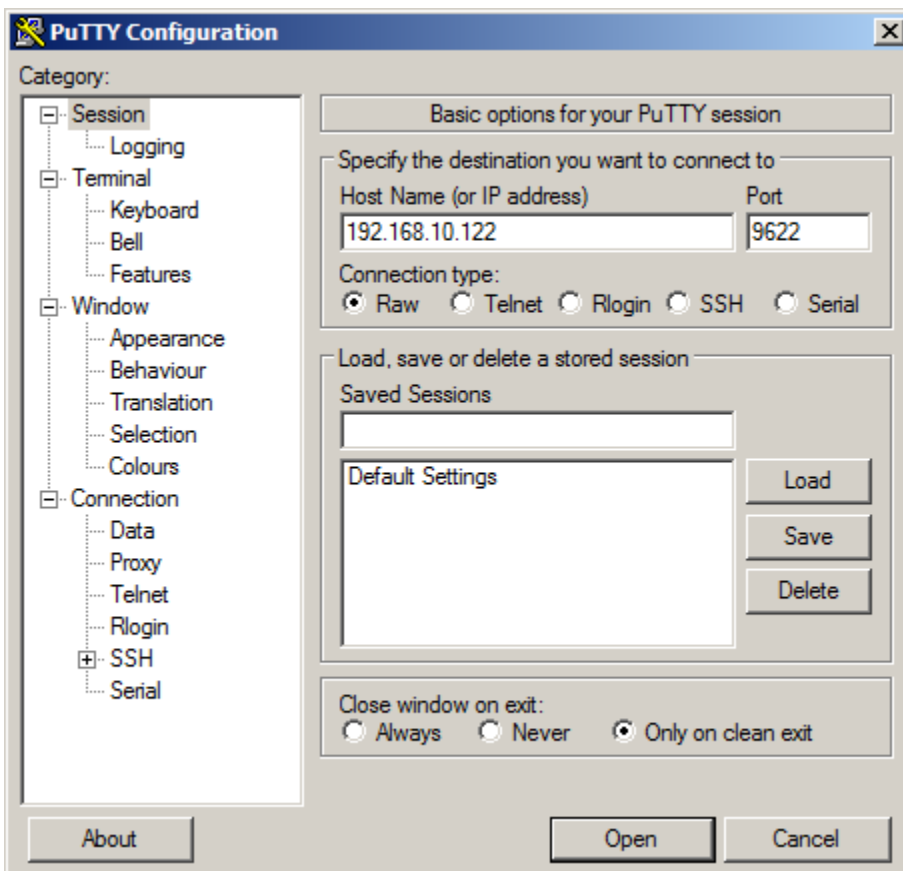


Table of Numeric Key Codes

Key Code	Val	Key Code	Val	Key Code	Val
1	1	Menu Down	34	Dim	67
2	2	Menu Left	35	Close	68
3	3	Menu Right	36	Open	69
4	4	Select	37	Stop 2	70
5	5	Exit	38	AM FM	71
6	6	Display	39	Cue	72
7	7	Guide	40	Disc Up	73
8	8	Page Up	41	Disc Down	74
9	9	Page Down	42	Info	75
0	10	Disc	43	External Src	76
Volume Up	11	Plus 10	44	Live/Intro	77
Volume Down	12	Open Close	45	Setup Menu	78
Mute	13	Random	46	Back	79
Chan Up	14	Track Fwd	47	Fav Channel	80
Chan Down	15	Track Rev	48	Display Fmt	81
Power	16	Sur On Off	49	SAP	82
Enter	17	Sur Mode	50	Slow	83
Prev Chan	18	Sur Up	51	PIP On	84
TV Video	19	Sur Down	52	PIP Off	85
TV VCR	20	PIP	53	PIP Freeze	86
A B	21	PIP Move	54	PIP Input	87
TV DVD	22	PIP Swap	55	PIP Chan Up	88
TV LD	23	Program	56	PIP Chan Dn	89
Input	24	Sleep	57	Input 1	90
TV DSS	25	On	58	Input 2	91
Play	26	Off	59	Input 3	92
Stop	27	11 / Red Button	60	Input 4	93

Search Fwd	28	12 / Yellow Button	61	Input 5	94
Search Rev	29	13 / Green Button	62	Input 6	95
Pause	30	14 / Blue Button	63	Input 7	96
Record	31	15 / DVR List	64	Input 8	97
Menu	32	16 / CC	65	Input 9	98
Menu Up	33	Bright	66	Input 10	99